

A Numerical Scheme for Viscous Ice Sheet Flow with a Dynamic Grounding Line

Charlie Ardern

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Supervised by Dr. Andrew Wells

Abstract

This project develops and tests a numerical scheme for the time evolution of a viscous ice sheet with a dynamic grounding line over a spatially varying basal topography. The moving-boundary problem is mapped onto a fixed interval, and a semi-implicit scheme is developed to solve the resulting system. The advective term is treated with upwinding, while the diffusive term is handled with a finite-volume discretisation. The floating ice shelf is then reconstructed from the grounding line history. Convergence tests demonstrate first-order accuracy in time and second-order accuracy in space over the parameter range considered. An empirical stability study indicates an approximately linear relationship between the time step and grid spacing, reminiscent of the CFL condition for the linear advection equation. Comparison with the analytical steady state shows that both the sheet profile and grounding line position approach the expected solution, demonstrating that the scheme is accurate and robust for this idealised model.

1 Introduction

Understanding the evolution of ice sheets is an important problem in climate science because it helps inform predictions of sea-level rise [1]. Of particular interest is the grounding line, the location at which the ice detaches from the bed and begins to form a floating ice shelf. This is important because it marks the transition from grounded, shear-dominated flow to floating extensional flow, so a shift in its position can significantly alter the rate at which ice is discharged into the ocean [2].

From a modelling perspective, this makes the ice sheet evolution a challenging moving-boundary problem, since the shape of the sheet changes over time and the position of the grounding line is not known in advance.

Simplified models of grounding line dynamics have been developed to study the coupled evolution of grounded ice and floating shelves. This project uses a model adapted from Robison et al. [3] to include a constant accumulation rate and spatially varying basal topography, allowing for a more realistic description of the system. We study an idealised two-dimensional ice sheet lying on a basal slope that varies in the downslope plane but remains constant in the third dimension. Ice is supplied in two ways: first, by constant accumulation due to snowfall directly onto the sheet, and second, by inflow entering at the top of the slope due to snowfall further upstream.

Although ice is commonly thought of as a solid, over sufficiently long timescales it deforms continuously and behaves as a highly viscous incompressible fluid [4]. Since the horizontal extent of the ice sheet far exceeds its thickness, the system can be approximated using a thin-film description. While the general behaviour of ice is non-Newtonian [5], the analysis in [3] adopts a Newtonian description for clarity and ease of modelling.

The floating shelf is assumed to remain in local hydrostatic balance and can be entirely reconstructed from the history of quantities at the grounding line. Since the primary focus of the project is the sheet evolution, this is intended purely as an illustration to provide a complete physical picture of the system.

Due to the high viscosity and moving boundary, the system is governed by a nonlinear PDE, so the general problem does not yield a simple analytical solution [3]. The key objective of this project is therefore to construct a numerical scheme to efficiently determine the time evolution of this system. This is achieved by transforming to dimensionless coordinates and mapping the domain onto an interval of fixed length. This introduces an effective advective term, while the original viscous flow produces a nonlinear diffusive term, motivating a discretisation that treats the two contributions separately. The report outlines the model in §2, discusses the numerical scheme in §3, presents convergence tests and stability studies in §4, and provides conclusions in §5. The scheme successfully achieves second-order spatial convergence and recovers the expected steady-state behaviours.

2 Ice Sheet Dynamics

The general setup of the system is shown in Figure 1. Ice of density ρ enters the system through accumulation a via snowfall on top of the ice sheet and from upstream flow through the boundary at $x = 0$. The ice is assumed to start at some distance L upstream, and flow down the bedrock $z = -b(x)$ towards the water, where buoyant forces cause it to break away at the grounding line $x = x_G(t)$, forming a shelf. Over time, the grounding line position x_G and the surface elevation h of the sheet will evolve as ice enters the system. The physical problem is therefore to determine the coupled quantities $x_G(t)$ and $h(x, t)$. After finding the sheet evolution, we can reconstruct the shelf from the grounding line history.

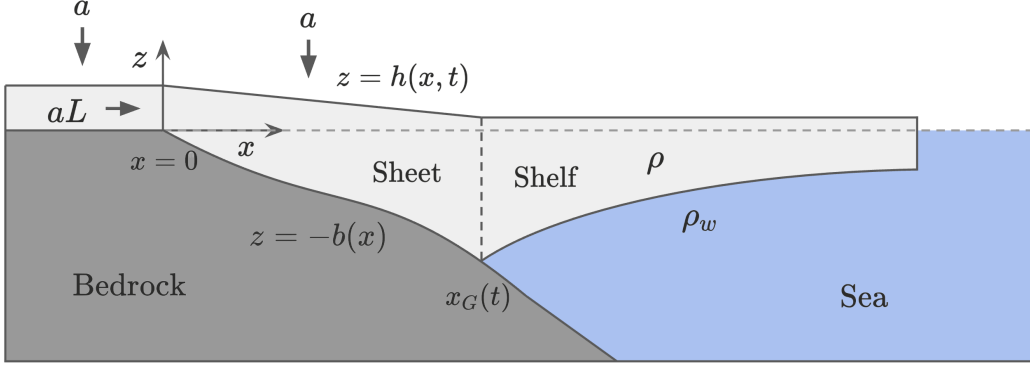


Figure 1: Overall setup for the ice sheet model. The line $z = 0$ represents the water level. The ice flux enters from negative x and flows towards positive x (left to right).

2.1 Ice sheet

The ice sheet is modelled as a highly viscous, incompressible Newtonian thin film of density ρ and dynamic viscosity η , positioned on a slope $z = -b(x)$. Over long timescales, the bed topography may evolve through sliding-driven subglacial erosion [6], but this effect is beyond the scope of our analysis. The surface elevation is modelled by $z = h(x, t)$ on the domain $x \in [0, x_G]$, and the system is assumed to be uniform over a large distance $w \gg x_G$ in the y direction. We assume there is a constant accumulation rate of a per unit area on top of the sheet, which leads to a constant positive volume flux wLa through the boundary at $x = 0$, where L is the upstream extent of the sheet in the negative x direction. The key steps leading to the governing equation are outlined here, with the full derivation in Appendix A.

Assuming a no-slip condition at $z = -b$ results in a viscous gravity current [7] with a velocity profile that varies quadratically with depth. Integrating this velocity over the thickness gives the volume flux:

$$q(x) = -\frac{w\rho g}{\eta} \frac{(h+b)^3}{3} \frac{\partial h}{\partial x}. \quad (2.1)$$

We can substitute this expression into the mass conservation equation for a slice of sheet with infinitesimal width dx :

$$\frac{\partial h}{\partial t} + \frac{1}{w} \frac{\partial q}{\partial x} - a = 0, \quad (2.2)$$

to arrive at the nonlinear, inhomogeneous diffusion equation governing $h(x, t)$:

$$\frac{\partial h}{\partial t} = \frac{\rho g}{3\eta} \frac{\partial}{\partial x} \left[(h+b)^3 \frac{\partial h}{\partial x} \right] + a. \quad (2.3)$$

The constant flux $q_0 = wLa$ at $x = 0$ leads to the first boundary condition:

$$-\frac{\rho g}{3\eta} [h(0) + b(0)]^3 \frac{\partial h}{\partial x} \Big|_{x=0} = La. \quad (2.4)$$

At the boundary $x = x_G$, hydrostatic balance in water of density ρ_w implies:

$$h(x_G, t) = \frac{\rho_w - \rho}{\rho} b(x_G), \quad (2.5)$$

which is the second boundary condition.

2.2 The Grounding Line

A fundamental assumption in viscous ice sheet modelling is that the depth-integrated longitudinal stress should balance on the two sides of the grounding line [2]. Following the derivation of [3, Sec. 2.3], this assumption leads to the following expression for the velocity of the grounding line:

$$\dot{x}_G = (h_G + b_G)^2 \left[\frac{g}{2\nu} \left(\frac{\partial h_G}{\partial x} \right)^2 - \frac{g'}{8\nu} \right] \left(\frac{\rho_w - \rho}{\rho} \frac{\partial b_G}{\partial x} - \frac{\partial h_G}{\partial x} \right)^{-1}, \quad (2.6)$$

where $\nu = \eta/\rho$, $h_G = h(x_G)$, $b_G = b(x_G)$, and $g' = g(\rho_w - \rho)/\rho_w$. However, in the case where the basal slope is steep, this velocity may be larger than the mean flow speed of the incoming ice, determined by the flux (2.1). In this case, the velocity of the grounding line is restricted to:

$$\dot{x}_G = \frac{q(x_G)}{w(h_G + b_G)} = -\frac{g(h_G + b_G)^2}{3\nu} \frac{\partial h_G}{\partial x}. \quad (2.7)$$

These two results can be combined to give the final expression for the velocity of the grounding line:

$$\dot{x}_G = \min \left[(2.6), (2.7) \right]. \quad (2.8)$$

2.3 Transforming the Sheet Equations

Before solving the system, it is advantageous to transform (2.3)–(2.5), and (2.8) to the dimensionless variables adapted from [3, Sec. 3.1]:

$$h = BC\hat{h}, \quad b = BC\hat{b}, \quad x = 2BC^4\hat{x}, \quad aLt = 2B^2C^5\hat{t},$$

with the constants:

$$B = (6\nu aL/g)^{1/3}, \quad C = (g/g')^{1/6}, \quad D = 2BC^4/L.$$

This reduces the dimensionless variables to be of order 1, allowing for easier computation and interpretation. We must also address the moving boundary at $x = x_G(t)$. A final transformation of variables $\chi = x/x_G(t)$ is made, mapping the domain onto a fixed interval $\chi \in [0, 1]$. Combining these transformations results in the following system:

$$\frac{\partial h}{\partial t} = \frac{\chi \dot{x}_G}{x_G} \frac{\partial h}{\partial \chi} + \frac{1}{x_G^2} \frac{\partial}{\partial \chi} \left[(h + b)^3 \frac{\partial h}{\partial \chi} \right] + D, \quad (2.9)$$

$$\left. \frac{(h + b)^3}{x_G} \frac{\partial h}{\partial \chi} \right|_{\chi=0} = -1, \quad (2.10)$$

$$h(\chi = 1) = \frac{\rho_w - \rho}{\rho} b_G, \quad (2.11)$$

$$\dot{x}_G = \min \left\{ \begin{aligned} & \frac{3}{2} \left(\frac{\rho_w b_G}{\rho} \right)^2 \frac{1}{x_G} \left[\left(\frac{\partial h_G}{\partial \chi} \right)^2 - x_G^2 \right] \left(\frac{\rho_w - \rho}{\rho} \frac{\partial b_G}{\partial \chi} - \frac{\partial h_G}{\partial \chi} \right)^{-1} \\ & - \left(\frac{\rho_w b_G}{\rho} \right)^2 \frac{1}{x_G} \frac{\partial h_G}{\partial \chi} \end{aligned} \right\} \quad (\chi = 1), \quad (2.12)$$

where the hats have been dropped for readability. After solving these equations, we can transform back to the real, dimensional variables to produce solutions for $h(x, t)$ and $\dot{x}_G$. With the exception of Section 2.4 and Appendix C, all variables written from this point onwards are given in their dimensionless form.

2.4 Ice shelf

The floating shelf is assumed to remain in local hydrostatic balance, so the total surface elevation h above sea level is related to the thickness H through $h = (\rho_w - \rho)H/\rho_w$. In addition, shear stresses above and below the shelf are neglected, leading to the approximation that the horizontal velocity is constant over the depth, so the motion is dominated by longitudinal stretching and vertical thinning. Following the derivation provided in [3, Sec. 2.1], we obtain the extensional flow equation:

$$\frac{\partial v_x}{\partial x} = \frac{g'}{8\nu} H. \quad (2.13)$$

This can be integrated twice to yield the following solution for the position of the slice of shelf which left the grounding line at $t = t_G$:

$$x(t, t_G) = \underbrace{x_G(t_G)}_{\text{original position}} + \underbrace{\int_{t_G}^t \frac{q_G(\tau)}{wH_G(\tau)} d\tau}_{\text{translation by incoming flow}} + \underbrace{\frac{g'}{8\nu w} \int_{t_G}^t (t - \tau) (q_G(\tau) - wH_G(\tau)\dot{x}_G(\tau)) d\tau}_{\text{displacement from extensional stretching of the shelf}}, \quad (2.14)$$

where x_G , H_G , and q_G represent the x -coordinate, thickness, and volume flux of the parcel of ice passing through the grounding line at the time of evaluation. Further, (2.13) can be substituted into the mass conservation equation (2.2) with $a = 0$ and $q = wv_x H$, and integrated to yield the thickness H at time t of this slice:

$$H(t, t_G) = \frac{8\nu H_G}{8\nu + g'H_G(t - t_G)}. \quad (2.15)$$

To compute the shelf, the quantities $x_G(t)$, $\dot{x}_G(t)$, $q_G(t)$, and $H_G(t)$ are found by solving (2.9)–(2.12). Subsequently, the position and thickness of each slice of shelf is found by evaluating (2.14) and (2.15), using numerical integration where required. The implementation details are provided in Appendix C.

3 Numerical Solution

Due to the nonlinear nature of this problem, there is no closed-form analytical solution for $h(x, t)$ in the general case. This requires the computation of an approximate solution to (2.9)–(2.12) using numerical methods. We achieve this by solving the system on a grid of N cells with spacing $\Delta\chi$ (see Figure 2), and determining an approximate expression for $\partial_t \mathbf{h}$ at each point in time, where $\mathbf{h} = [h_0, h_1, \dots, h_{N-1}] \in \mathbb{R}^N$ are the h values at the centre of each cell. Next, we take time steps using an update rule:

$$\mathbf{h}^{n+1} = \mathbf{f}^n(\mathbf{h}^n, \Delta t, \Delta\chi), \quad (3.1)$$

where $\mathbf{h}^n$ denotes the set of h values at time $t = n\Delta t$ with some finite time step Δt . The form of $\mathbf{f}^n$ will be derived in the following sections.

The equation to be solved (2.9) contains an advective term, a diffusive term, and a constant term. The following sections outline the numerical procedures for approximating them.

3.1 The Advective Term

The first term in the right-hand side of (2.9) is approximated using an upwind scheme [8], where the derivatives are taken upstream to respect the direction of the information flow and produce a more stable solution. For the case where $\dot{x}_G < 0$, information flows in the positive χ direction, so the derivative is taken backwards:

$$\left. \frac{\partial h_i}{\partial t} \right|_{\text{adv}} \equiv \chi \frac{\dot{x}_G}{x_G} \frac{\partial h}{\partial \chi} \approx \chi_i \frac{\dot{x}_G}{x_G} \frac{3h_i - 4h_{i-1} + h_{i-2}}{2\Delta\chi}. \quad (3.2)$$

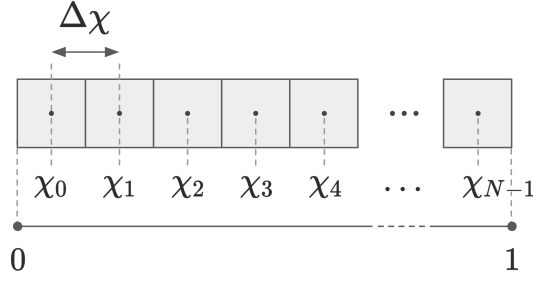


Figure 2: The grid notation convention used, with cell i centred at $\chi_i = (\frac{1}{2} + i)\Delta\chi$. The left boundary is positioned at $\chi_{-\frac{1}{2}} = 0$ and the right boundary is positioned at $\chi_{N-\frac{1}{2}} = 1$.

In the case where $\dot{x}_G > 0$, information flows in the negative χ direction, so the derivative is taken forwards:

$$\left. \frac{\partial h_i}{\partial t} \right|_{\text{adv}} \approx \chi_i \frac{\dot{x}_G}{x_G} \frac{-3h_i + 4h_{i+1} - h_{i+2}}{2\Delta\chi}. \quad (3.3)$$

Finally, in the case where $\dot{x}_G = 0$, the whole term goes to zero, so no derivative is required. In both (3.2) and (3.3), a three-point stencil is used, giving a truncation error of $\mathcal{O}(\Delta\chi^2)$.

3.2 The Diffusive Term

The second term in (2.9) is approximated using a finite-volume method [9], which is locally conservative by construction, since it is obtained by integrating a conservation law over each cell and balancing numerical fluxes across cell faces. First, we define the flux as:

$$F(\chi) \equiv (h + b)^3 \frac{\partial h}{\partial \chi}, \quad (3.4)$$

which can be substituted into the diffusive term to arrive at the standard form of a conservation law:

$$\left. \frac{\partial h}{\partial t} \right|_{\text{diff}} \equiv \frac{1}{x_G^2} \frac{\partial F}{\partial \chi}. \quad (3.5)$$

This equation is then averaged over a cell of width $\Delta\chi$:

$$\frac{1}{\Delta\chi} \int_{\chi_{i-\frac{1}{2}}}^{\chi_{i+\frac{1}{2}}} \left[\left. \frac{\partial h}{\partial t} \right|_{\text{diff}} \right] d\chi = \frac{1}{\Delta\chi} \int_{\chi_{i-\frac{1}{2}}}^{\chi_{i+\frac{1}{2}}} \frac{1}{x_G^2} \frac{\partial F}{\partial \chi} d\chi,$$

leading to the following approximation for (3.5) in terms of the fluxes on the boundaries:

$$\left. \frac{\partial h_i}{\partial t} \right|_{\text{diff}} \approx \frac{1}{x_G^2} \frac{F_{i+\frac{1}{2}} - F_{i-\frac{1}{2}}}{\Delta\chi}, \quad (3.6)$$

with $F_i = F(\chi_i)$ and $h_i = h(\chi_i)$. The expression for $F_{i+\frac{1}{2}}$ is given by:

$$F_{i+\frac{1}{2}} \equiv \left(\frac{h_{i+1} + h_i}{2} + b_{i+\frac{1}{2}} \right)^3 \left(\frac{h_{i+1} - h_i}{\Delta\chi} \right) \equiv K_{i+\frac{1}{2}} \left(\frac{h_{i+1} - h_i}{\Delta\chi} \right), \quad (3.7)$$

where $h_{i\pm\frac{1}{2}}$ and $\partial_\chi h_{i\pm\frac{1}{2}}$ have been approximated with second-order centred differences, and $b_i = b(\chi_i)$. The expression for $F_{i-\frac{1}{2}}$ follows trivially by decrementing the index. The final expression (3.6) has a truncation error of $\mathcal{O}(\Delta\chi^2)$.

3.3 Full Numerical Scheme

The diffusive term contains both a second derivative and a variable coefficient, which makes it highly unstable when used with an explicit scheme, where the time derivative at step n is calculated entirely from the system variables at n . The most stable case is to use an implicit solver, where the time derivative at n is calculated from quantities at time step $n + 1$. However, for our system, K would depend on the unknown h^{n+1} through $(h^{n+1} + b)^3$, so a fully implicit scheme would involve running a nonlinear solve at each time step which is computationally expensive. We overcome this issue by using a semi-implicit scheme, where the derivatives in the flux terms (3.7) are computed at time step $n + 1$, while the rest of the equation is computed at step n . Using (3.6) and (3.7), this can be written as:

$$\frac{h_i^{n+1} - h_i^n}{\Delta t} = \frac{\partial h_i}{\partial t} \Big|_{\text{adv}}^n + \frac{1}{(x_G^n)^2 \Delta \chi} \left[K_{i+\frac{1}{2}}^n \left(\frac{h_{i+1}^{n+1} - h_i^{n+1}}{\Delta \chi} \right) - K_{i-\frac{1}{2}}^n \left(\frac{h_i^{n+1} - h_{i-1}^{n+1}}{\Delta \chi} \right) \right] + D. \quad (3.8)$$

This system can be solved by setting up a matrix $\mathbf{M}^n$ such that:

$$\mathbf{M}^n \mathbf{h}^{n+1} = \mathbf{g}^n, \quad (3.9)$$

where $\mathbf{g}^n \in \mathbb{R}^N$ is a linear function of $\mathbf{h}^n$. More detail can be found in Appendix B. It is clear from the left-hand side of (3.8) that the scheme is first-order in time, with a truncation error of $\mathcal{O}(\Delta t)$.

3.4 Boundary Conditions

We conclude our discussion of the numerical scheme with an explanation of the boundary conditions. Condition (2.10) at $\chi = 0$ simply states:

$$F_{-\frac{1}{2}} = -x_G. \quad (3.10)$$

Next, the condition (2.11) can be enforced by setting $h_{N-\frac{1}{2}} = h_G = (\rho_w - \rho)b_G/\rho$, and fitting a quadratic function of χ to $h_{N-\frac{1}{2}}$, h_{N-1} , and h_{N-2} to define the ghost cell:

$$h_N = \frac{1}{3}(8h_G - 6h_{N-1} + h_{N-2}), \quad (3.11)$$

which can be used to approximate the derivative at the boundary $\chi = 1$:

$$\frac{\partial h_{N-1/2}}{\partial \chi} \approx \frac{8h_G + h_{N-2} - 9h_{N-1}}{3\Delta \chi}. \quad (3.12)$$

Finally, the expression for $\dot{x}_G$ is found by substituting (3.12) into (2.12).

To compute the advective term (3.2), we must find expressions for the derivatives at the boundaries. For $\partial_\chi h_0$ and $\partial_\chi h_1$ in (3.2), we define the ghost cell h_{-1} from the flux condition (2.10):

$$\left(\frac{h_0 + h_{-1}}{2} + b_{-\frac{1}{2}} \right)^3 \left(\frac{h_0 - h_{-1}}{\Delta \chi} \right) = -x_G, \quad (3.13)$$

and find h_{-1} using a numerical root-finding algorithm. The derivatives at and adjacent to the boundary can then be approximated as:

$$\frac{\partial h_0}{\partial \chi} = \frac{h_1 - h_{-1}}{2\Delta \chi}, \quad \frac{\partial h_1}{\partial \chi} = \frac{3h_1 - 4h_0 + h_{-1}}{2\Delta \chi}. \quad (3.14)$$

For $\partial_\chi h_{N-1}$ and $\partial_\chi h_{N-2}$ in (3.3), the ghost cell (3.11) can be substituted into:

$$\frac{\partial h_{N-1}}{\partial \chi} = \frac{h_N - h_{N-2}}{2\Delta \chi}, \quad \frac{\partial h_{N-2}}{\partial \chi} = \frac{-3h_{N-2} + 4h_{N-1} - h_N}{2\Delta \chi}. \quad (3.15)$$

We have now developed a scheme in which the boundary values are second-order accurate, leading to an overall truncation error of $\mathcal{O}(\Delta\chi^2)$.

The scheme was implemented in the Python programming language, making use of the NumPy linear algebra library [10] for high-speed vectorised operations and the SciPy scientific computing library [11] for efficient solves in (3.9) and (3.13).

4 Numerical Tests

The purpose of this section is to assess the performance, stability, and accuracy of the scheme. Unless stated otherwise, the system was evolved from an initial state of $x_G(0) = 0.1$ and $h(x, 0) = 0.7(1 - 0.9x)$ for ease of visualisation. The choice to set $x_G(0) \neq 0$ ensures the $1/x_G$ factors remain finite.

4.1 General Behaviour

An example setup is shown in Figure 3, where the sheet and shelf have been plotted at several dimensionless times using a basal slope $b(x) = 0.2\sin(8x) + 1.3x$ and an accumulation rate $D = 0.4$. A simple visual inspection shows the grounding line advancing down the slope over time, before reaching a steady position. As expected from the continuous flux of ice, the shelf extends from the bed and increases in length without limit, while the sheet profile approaches a steady state.

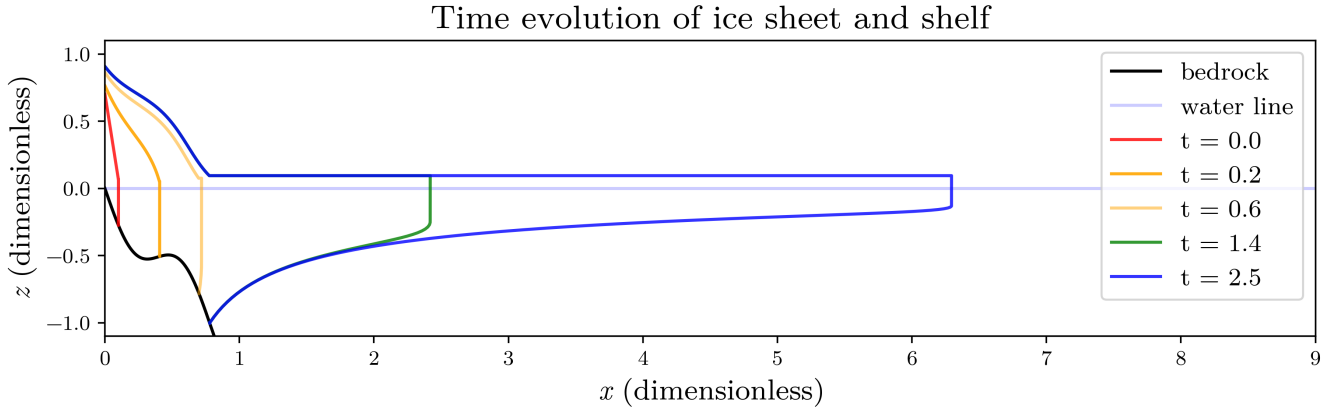


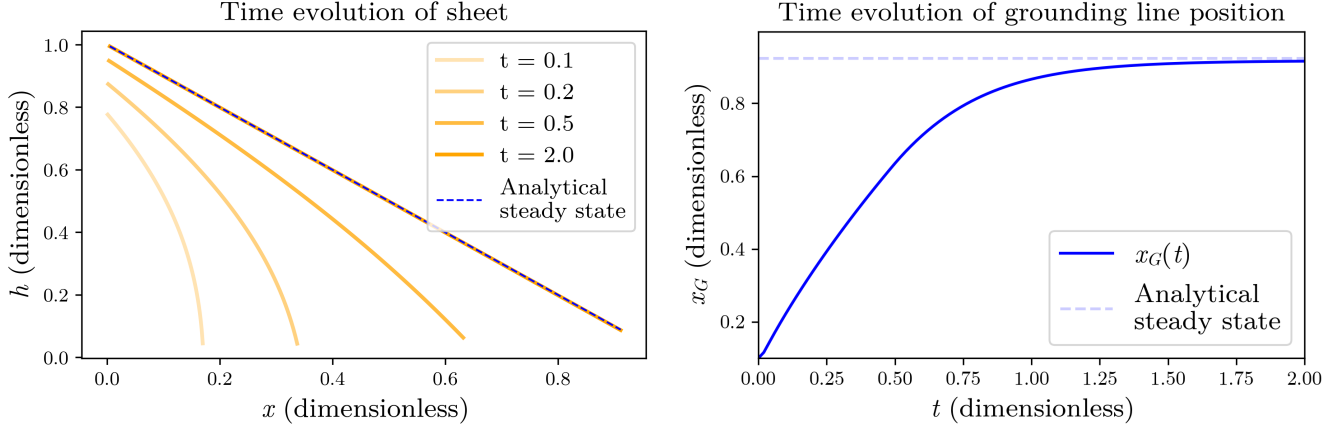
Figure 3: The form of the sheet and shelf at several dimensionless times.

While no analytical solution exists for the general problem, steady-state solutions can be found for certain cases, and are an effective method for verifying the validity of a numerical scheme. For the sheet, the full functional form of the steady state can be found [3] when the basal slope is linear and $D = 0$. Using $b(\chi) = A\chi x_G$ with $A \in \mathbb{R}$ and setting $\dot{x}_G = \dot{h} = 0$ in the system (2.9)–(2.12) yields:

$$(h + Ax_G\chi)^3 \frac{\partial h}{\partial \chi} = -x_G, \quad h_{\chi=1} = \epsilon x_G, \quad \left. \frac{\partial h}{\partial \chi} \right|_{\chi=1} = -x_G,$$

with $\epsilon = \rho_w g' A / \rho g$ which can be combined to yield a steady-state value of $x_G = (A + \epsilon)^{-1}$. For the trivial case where $A = 1$, the steady-state solution for $h(\chi)$ can be written as $h(\chi) = 1 - x_G\chi$, or equivalently $h(x) = 1 - x$. This scenario was simulated and the profile of the sheet and position of the grounding line have been plotted over time in Figures 4a and 4b, respectively, with both showing a clear convergence towards the expected analytical solution.

It should be noted that the purpose of this test is to qualitatively verify that the solution behaves as expected in known cases. Numerical schemes can be used to compute solutions to an arbitrarily small error given sufficient refinement, an effect that is explored in the next section.

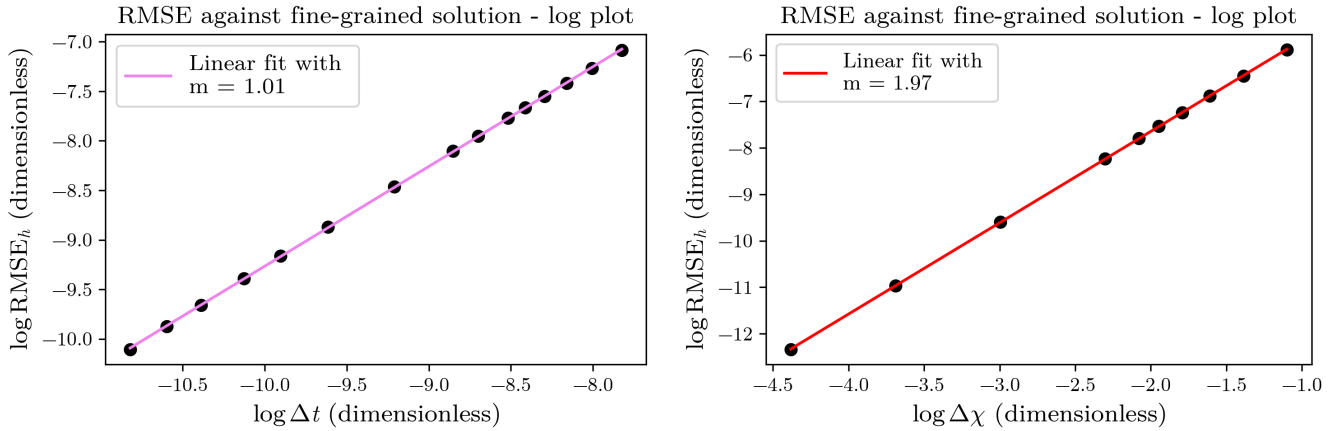


(a) The sheet surface elevation at four different times compared with the true steady-state shape. (b) The grounding line position against time compared with the true steady-state value $x_G = 0.917$.

Figure 4: Comparison of the sheet and grounding line time evolution with the analytical steady state $h(x) = 1 - x$, for the case where $D = 0$. The numerical solution approaches the analytical steady state as time progresses.

4.2 Temporal Convergence

The time derivatives in the scheme are taken to be first-order approximations, leading to an expected time-discretisation error of order $\mathcal{O}(\Delta t)$. A fine-grained solution was computed with a dimensionless time step of $\Delta t = 1 \times 10^{-6}$ and used as a reference solution against solutions with a range of larger time steps. A fine grid size of $\Delta \chi = 6.7 \times 10^{-4}$ was used to minimise the spatial discretisation error $\mathcal{O}(\Delta \chi^2)$, ensuring that the time error dominates. For each time step size, the root-mean-square error was computed against the reference solution. A logarithmic plot of the results is shown in Figure 5a at $t = 0.2$ for a bed of slope $b(x) = 0.8x$ and an accumulation rate $D = 0.4$. A linear model was fitted, yielding a gradient of $1.01 \approx 1$, consistent with the predicted linear order¹ $\mathcal{O}(\Delta t)$.



(a) $\log(\text{RMSE})$ against $\log(\Delta t)$, computed relative to a fine-grained solution, with a linear fit. (b) $\log(\text{RMSE})$ against $\log(\Delta \chi)$, computed relative to a fine-grained solution, with a linear fit.

Figure 5: Temporal and spatial convergence of the numerical scheme relative to fine-grained reference solutions. Panel (a) shows first-order convergence in time, while panel (b) shows second-order convergence in space.

¹For a relationship $y = ax^n$, taking the logarithm yields $\log y = n \log x + \log a$, so the gradient of the plot is n .

4.3 Spatial Convergence

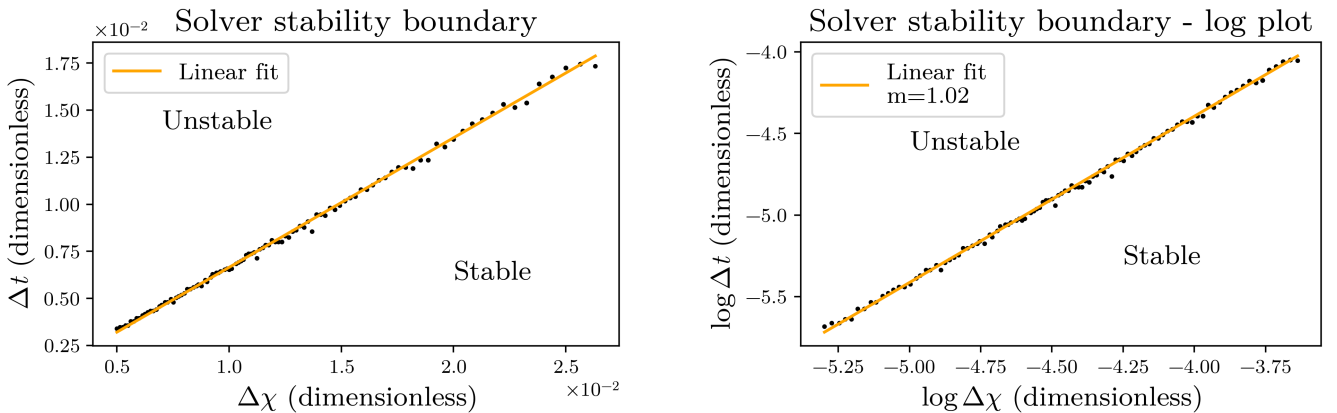
In analogy with the temporal convergence test, a fine-grained solution ($\Delta\chi = 1.4 \times 10^{-3}$) was used as a reference solution for a range of solves with larger spacings. Each was run with a small dimensionless time step of $\Delta t = 2.7 \times 10^{-6}$ to ensure that the discretisation error for space dominated that for time. The solutions were interpolated onto the reference grid², and the root-mean-square error was computed against the reference solution. The system was evaluated at a dimensionless time $t = 0.8$, with a basal slope $b(x) = 2x$, initial grounding line position $x_G(0) = 0.5$, and an accumulation rate $D = 0.4$. A logarithmic plot of the results is shown in Figure 5b. A linear model was fitted, yielding a gradient of $1.97 \approx 2$, which is consistent with the predicted quadratic order $\mathcal{O}(\Delta\chi^2)$.

4.4 Stability

In this section, we will explore the stability of the scheme. For linear PDEs, the stability criteria of a numerical scheme can often be found through techniques such as von Neumann stability analysis [12], yielding simple relationships between Δt and $\Delta\chi$. While no such simple closed-form expression exists for the nonlinear system in this study, a search can be performed to find the maximum time step for which a solution is stable for a range of different $\Delta\chi$.

To check for stability, the code first checked whether any ‘NaN’ values were present in the output. If there were none, the RMSE for the proposed solution was computed against a reference solution which was known to converge, providing a coarse indication of whether the numerical profile remained on the same qualitative solution branch. Errors below 0.05 were classified as stable, while larger values were deemed to have profiles that deviated into non-physical shapes. For the parameters tested, the RMSE values of the stable cases were found to be approximately 0.01 or less. This made 0.05 a sensible decision boundary that ensured stable parameters were found while minimising the misclassification of unstable cases. The search was carried out on a 200×200 grid in $(\Delta\chi, \Delta t)$ space with $\Delta t \in [0.0020, 0.0175]$ and $\Delta\chi \in [0.005, 0.030]$, with the results presented in Figure 6a. The evident linearity of the boundary is verified in Figure 6b, which shows the boundary in $(\log \Delta\chi, \log \Delta t)$ space. A linear model was fitted with a gradient of $1.02 \approx 1$, consistent with a linear relationship between $\Delta\chi$ and Δt .

This approximately linear relationship is reminiscent of the CFL condition for linear advective PDEs, in which $u\Delta t \leq \Delta\chi$ is required for stability, with u constant. In the present scheme, the semi-implicit treatment of the diffusive term greatly relaxes the corresponding time step restriction, so the dominant stability constraint arises from the explicitly treated advective term. Therefore, it is reasonable that the empirical stability boundary is close to linear in Δt and $\Delta\chi$, and over the parameter range tested, is broadly advective in character.



(a) The approximate boundary in $(\Delta\chi, \Delta t)$ space at which the scheme becomes unstable.

(b) The approximate stability boundary with a linear fit in $(\log \Delta\chi, \log \Delta t)$ space.

Figure 6: The empirical stability boundary computed by a search over a 200×200 grid in $(\Delta\chi, \Delta t)$ space.

²Linear interpolation also has an error of order $\mathcal{O}(\Delta\chi^2)$, but this is with respect to the fine-grained grid spacing. It is therefore negligible compared with the spatial discretisation error of the coarse grids.

5 Conclusion

This project developed and tested a numerical scheme for a nonlinear model of viscous ice sheet flow with a moving grounding line and freely floating ice shelf. The equations were first transformed to dimensionless variables before mapping the moving domain onto an interval of constant width. The resulting PDE was solved using a semi-implicit scheme, with upwinding used for the advective term and a finite-volume treatment of the diffusive term. This provided an efficient way to compute the grounded sheet and reconstruct the shelf from the grounding line history.

The numerical tests demonstrated that the scheme reproduced the expected behaviour of the system. In the steady-state test, both the sheet profile and grounding line position were found to approach the analytical solution. The spatial and temporal convergence studies showed a first-order error in time, and a second-order error in space, consistent with the truncation errors of the scheme. This provides further confidence that the discretisation was implemented correctly. The empirical stability study identified an approximate linear stability requirement between the time step and grid spacing, suggesting that the dominant time step restriction arose from the explicitly treated advective term. This agrees with the intuition that semi-implicit treatment suppressed the time step restriction from the diffusive term.

Overall, the method developed here provides a practical and robust framework for modelling viscous ice sheet dynamics over spatially varying basal topographies with accumulation. Although the model is idealised and relies on several simplifying assumptions, the results suggest that the scheme is accurate and computationally effective over the parameter range considered. Natural extensions would include melting, time-dependent basal topography, and the addition of a third dimension, bringing the model closer to real ice sheet applications.

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A Derivation of the Sheet Equation

This section follows the general method outlined in [7]. We start with the Navier–Stokes equation:

$$\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla) \mathbf{v} = -\frac{1}{\rho} \nabla p + \frac{\eta}{\rho} \nabla^2 \mathbf{v} - g \hat{\mathbf{z}}. \quad (\text{A.1})$$

For a thin film of thickness $h \ll x_G$ and velocity scale U satisfying $\rho h^2 U \ll \eta x_G$, this reduces to:

$$\frac{\partial p}{\partial x} = \eta \frac{\partial^2 v_x}{\partial z^2}, \quad \frac{\partial p}{\partial z} = -\rho g. \quad (\text{A.2})$$

Integrating the second equation with the boundary condition $p(h) = p_{\text{atm}}$ for constant atmospheric pressure p_{atm} gives:

$$p(z) = p_{\text{atm}} + \rho g(h - z). \quad (\text{A.3})$$

Substituting this into the first of (A.2) yields:

$$\rho g \frac{\partial h}{\partial x} = \eta \frac{\partial^2 v_x}{\partial z^2}, \quad (\text{A.4})$$

which can be integrated twice with respect to z to give:

$$v_x(z) = \frac{\rho g}{\eta} \frac{\partial h}{\partial x} \left[\frac{1}{2} z^2 - h z - \frac{1}{2} b^2 - h b \right], \quad (\text{A.5})$$

where we have used the boundary conditions $v_x(-b) = 0$ and $\frac{\partial v_x(h)}{\partial z} = 0$.

Next, we use the continuity equation:

$$\frac{\partial V}{\partial t} = - \oint_{\partial V} \mathbf{J} \cdot d\mathbf{S}, \quad (\text{A.6})$$

where $\mathbf{J}$ is the volume flux per unit area. By taking the region to be a slice of the sheet of infinitesimal width dx , this becomes:

$$\begin{aligned} \frac{\partial}{\partial t} (h w dx) &= q(x) - q(x + dx) + a w dx, \\ \implies \frac{\partial h}{\partial t} &= -\frac{1}{w} \frac{\partial q}{\partial x} + a. \end{aligned} \quad (\text{A.7})$$

The total volume flux $q(x)$ is found by integrating (A.5) over the height of the slice:

$$q(x) = w \int_{-b}^h v_x dz = -\frac{w \rho g}{\eta} \frac{(h + b)^3}{3} \frac{\partial h}{\partial x}. \quad (\text{A.8})$$

This can be substituted into (A.7) to arrive at the final result:

$$\frac{\partial h}{\partial t} = \frac{\partial}{\partial x} \left[\frac{\rho g}{\eta} \frac{(h + b)^3}{3} \frac{\partial h}{\partial x} \right] + a. \quad (\text{A.9})$$

B Full Derivation of the Numerical Scheme

This section details the vectorised implementation of the scheme which was used for efficient computation. The symbol $\odot$ denotes element-wise multiplication. Defining $N = 1/\Delta x$ as the total number of cells, the advective term is written in vectorised form as:

$$\left. \frac{\partial \mathbf{h}_i}{\partial t} \right|_{\text{adv}} = \frac{\dot{x}_G}{x_G} \mathbf{u}_1 \odot \mathbf{u}_2, \quad (\text{B.1})$$

with $\mathbf{u}_1 \in \mathbb{R}^N$ given by:

$$\mathbf{u}_1 = \left[\frac{1}{2}, \frac{3}{2}, \dots, i + \frac{1}{2}, \dots, N - \frac{3}{2}, N - \frac{1}{2} \right],$$

and $\mathbf{u}_2 \in \mathbb{R}^N$ given by:

$$\mathbf{u}_2 = \frac{1}{2} \left[h_1 - h_{-1}, 3h_1 - 4h_0 + h_{-1}, \dots, 3h_i - 4h_{i-1} + h_{i-2}, \dots, 3h_{N-1} - 4h_{N-2} + h_{N-3} \right],$$

for the case where $\dot{x}_G < 0$, and

$$\mathbf{u}_2 = \frac{1}{2} \left[-3h_0 + 4h_1 - h_2, -3h_1 + 4h_2 - h_3, \dots, -3h_i + 4h_{i+1} - h_{i+2}, \dots, -3h_{N-2} + 4h_{N-1} - h_N, h_N - h_{N-2} \right],$$

for the case where $\dot{x}_G > 0$. Here, h_{-1} is defined by (3.13) and h_N is defined by (3.11). Next, we approximate the diffusive term. We start by defining:

$$C^n = \frac{\Delta t}{(x_G^n)^2 \Delta \chi^2},$$

and substituting into (3.8) to yield the general update rule:

$$h_i^n + (A_i^n + D)\Delta t = - \underbrace{\left[C^n K_{i-\frac{1}{2}}^n \right]}_{M_i} h_{i-1}^{n+1} + \underbrace{\left[1 + C^n (K_{i+\frac{1}{2}}^n + K_{i-\frac{1}{2}}^n) \right]}_{M'_i} h_i^{n+1} - \underbrace{\left[C^n K_{i+\frac{1}{2}}^n \right]}_{M''_i} h_{i+1}^{n+1}. \quad (\text{B.2})$$

where A_i^n is the advective term given by (3.2)–(3.3) at time step n . These elements M_i , M'_i , M''_i are sufficient to construct all but the first and final rows of the matrix, where boundary conditions must be used. The first row is found by inserting $F_{-\frac{1}{2}}^{n,n+1} = -x_G$ into (3.8) in accordance with (3.10). This leads to:

$$h_0^n + (A_0^n + D)\Delta t + C^n x_G^n \Delta \chi = \underbrace{\left[1 + C^n K_{\frac{1}{2}}^n \right]}_{M'_0} h_0^{n+1} - \underbrace{\left[C^n K_{\frac{1}{2}}^n h_1^{n+1} \right]}_{M''_0}. \quad (\text{B.3})$$

Finally, using (3.12) to approximate the derivative in the term $F_{N-\frac{1}{2}}^{n,n+1}$ yields:

$$h_{N-1}^n + (A_{N-1}^n + D)\Delta t + 8C^n B^n h_G^n = - \underbrace{\left[C^n B^n + C^n K_{N-\frac{3}{2}}^n \right]}_{M_{N-1}} h_{N-2}^{n+1} + \underbrace{\left[1 + 9C^n B^n + C^n K_{N-\frac{3}{2}}^n \right]}_{M'_{N-1}} h_{N-1}^{n+1}, \quad (\text{B.4})$$

where we have defined:

$$B^n = \frac{1}{3} \left(\frac{\rho_w b_G^n}{\rho} \right)^3.$$

We can now construct the full matrix:

$$\underbrace{\begin{pmatrix} h_0^n \\ h_1^n \\ h_2^n \\ h_3^n \\ \vdots \\ h_{N-2}^n \\ h_{N-1}^n \end{pmatrix}}_{\mathbf{g}^n} + \underbrace{\begin{pmatrix} C^n x_G^n \Delta \chi \\ 0 \\ 0 \\ 0 \\ \vdots \\ 0 \\ 8C^n B^n h_G^n \end{pmatrix}}_{\mathbf{g}^n} + \underbrace{\begin{pmatrix} A_0^n + D \\ A_1^n + D \\ A_2^n + D \\ A_3^n + D \\ \vdots \\ A_{N-2}^n + D \\ A_{N-1}^n + D \end{pmatrix}}_{\mathbf{g}^n} \Delta t = \underbrace{\begin{pmatrix} M'_0 & M''_0 & 0 & 0 & \dots & 0 & 0 \\ M_1 & M'_1 & M''_1 & 0 & \dots & 0 & 0 \\ 0 & M_2 & M'_2 & M''_2 & \dots & 0 & 0 \\ 0 & 0 & M_3 & M'_3 & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \dots & M_{N-1} & M'_{N-1} \end{pmatrix}}_{\mathbf{M}^n} \underbrace{\begin{pmatrix} h_0^{n+1} \\ h_1^{n+1} \\ h_2^{n+1} \\ h_3^{n+1} \\ \vdots \\ h_{N-2}^{n+1} \\ h_{N-1}^{n+1} \end{pmatrix}}_{\mathbf{h}^{n+1}}, \quad (\text{B.5})$$

this tridiagonal matrix system is then solved using the Python package SciPy [13].

C Computation of the Shelf

We start by introducing a set of vectors $\mathbf{x}_G, \dot{\mathbf{x}}_G, \mathbf{t}_G, \mathbf{H}_G \in \mathbb{R}^T$ which store the respective parameters at each of the T time steps. These are determined by solving the sheet system, (2.9)–(2.12), and converted back to dimensional variables before use. Next, the analogous flux vector $\mathbf{q}_G \in \mathbb{R}^T$ is found through:

$$\mathbf{q}_G = -\frac{wg}{3\nu} \mathbf{H}_G^3 \odot \frac{\partial \mathbf{h}_G}{\partial x}, \quad (\text{C.1})$$

with $\odot$ representing element-wise multiplication and $\partial \mathbf{h}_G / \partial x$ determined by transforming (3.12) back to the appropriate coordinates. Having determined these variables, we now introduce the following notation for ease of implementation. For a given $\mathbf{a} = (a_0, a_1, \dots, a_{N-1})$:

$$\mathbf{a}[i] = a_i, \quad \mathbf{a}[i:] = (a_i, a_{i+1}, \dots, a_{N-1}), \quad \text{sum}(\mathbf{a}) = \sum_{j=0}^{N-1} a_j.$$

This notation originates directly from the NumPy [10] package in the Python programming language. We also define $\oslash$ as element-wise division. We now write the numerical integration of (2.14) as:

$$\begin{aligned} \mathbf{x}[i] &= \mathbf{x}_G[i] + \frac{1}{w} \text{sum}\left((\mathbf{q}_G \oslash \mathbf{H}_G)[i:]\right) \Delta t + \frac{g'}{8\nu w} \text{sum}\left(\boldsymbol{\zeta}_1 \odot \boldsymbol{\zeta}_2\right) \Delta t, \\ \boldsymbol{\zeta}_1 &= t\mathbf{1} - \mathbf{t}_G[i:], \\ \boldsymbol{\zeta}_2 &= \mathbf{q}_G[i:] - w\mathbf{H}_G[i:] \odot \dot{\mathbf{x}}_G[i:], \end{aligned} \quad (\text{C.2})$$

where $\mathbf{1} = (1, 1, \dots, 1) \in \mathbb{R}^{(T+1-i)}$ and $\mathbf{x} \in \mathbb{R}^{t/\Delta t+1}$ is a vector whose i th component is the position of the shelf slice which left the grounding line at $t = i\Delta t$. As a result, the dimension of this vector increases each time step, as a new slice of shelf leaves the grounding line. The thickness of the shelf is computed by writing (2.15) in vectorised form:

$$\mathbf{H}(t) = 8\nu \mathbf{H}_G \oslash \left(g' \mathbf{H}_G \odot (t\mathbf{1} - \mathbf{t}_G) + 8\nu \mathbf{1}\right), \quad (\text{C.3})$$

where $\mathbf{H} \in \mathbb{R}^{t/\Delta t+1}$ is the thickness of the ice at each point in $\mathbf{x}$ at time t .